

-- =====

-- Phase 1 - Core ERP Foundation (REVISED) - PART 1 OF 3

-- Database Engine: PostgreSQL 14

-- =====

CREATE EXTENSION IF NOT EXISTS pgcrypto;

-- 1. erp_companies

```
CREATE TABLE erp_companies (  
    company_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL,  
    legal_name VARCHAR(255) NOT NULL,  
    display_name VARCHAR(255),  
    company_type SMALLINT NOT NULL DEFAULT 1,  
    national_id VARCHAR(50),  
    economic_code VARCHAR(50),  
    registration_number VARCHAR(100),  
    is_legal_entity BOOLEAN NOT NULL DEFAULT FALSE,  
    phone VARCHAR(50),  
    email VARCHAR(150),  
    address TEXT,  
    status SMALLINT NOT NULL DEFAULT 1,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    created_by UUID,  
    updated_at TIMESTAMPTZ,  
    updated_by UUID,
```

```

deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT \
);

CREATE UNIQUE INDEX uq_erp_companies_national_id ON erp_companies(tenant_id,
national_id) WHERE deleted_at IS NULL;

CREATE INDEX idx_erp_companies_tenant_id ON erp_companies(tenant_id) WHERE deleted_at
IS NULL;

```

-- ୨. erp_branches

```

CREATE TABLE erp_branches (
branch_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
company_id UUID NOT NULL REFERENCES erp_companies(company_id) ON DELETE RESTRICT,
parent_branch_id UUID REFERENCES erp_branches(branch_id) ON DELETE RESTRICT,
code VARCHAR(୦୦) NOT NULL,
name VARCHAR(୨୦୦) NOT NULL,
is_main_branch BOOLEAN NOT NULL DEFAULT FALSE,
phone VARCHAR(୦୦),
address TEXT,
status SMALLINT NOT NULL DEFAULT \,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT \

```

```
);
```

```
CREATE UNIQUE INDEX uq_branch_company_code ON erp_branches(company_id, code)
WHERE deleted_at IS NULL;
```

```
-- ۳. erp_departments
```

```
CREATE TABLE erp_departments (
    department_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    company_id UUID NOT NULL REFERENCES erp_companies(company_id) ON DELETE RESTRICT,
    branch_id UUID REFERENCES erp_branches(branch_id) ON DELETE RESTRICT,
    parent_department_id UUID REFERENCES erp_departments(department_id) ON DELETE
    RESTRICT,
    code VARCHAR(۵۰) NOT NULL,
    name VARCHAR(۲۰۰) NOT NULL,
    description VARCHAR(۵۰۰),
    status SMALLINT NOT NULL DEFAULT ۱,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT ۱
);
```

```
CREATE UNIQUE INDEX uq_department_company_code ON erp_departments(company_id,
code) WHERE deleted_at IS NULL;
```

```
-- ۴. users (کاربران فراتننتی عمومی پلتفرم)
```

```

CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    username VARCHAR(100),
    email VARCHAR(100),
    mobile VARCHAR(20),
    first_name VARCHAR(100),
    last_name VARCHAR(100),
    user_kind SMALLINT NOT NULL DEFAULT 1,
    status SMALLINT NOT NULL DEFAULT 1,
    last_login_at TIMESTAMPTZ,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT 1
);

CREATE INDEX idx_users_status ON users(status) WHERE deleted_at IS NULL;

```

```
-- =====
```

```
-- Phase 1 - Core ERP Foundation (REVISED) - PART 2 OF 4
```

```
-- Database Engine: PostgreSQL 14
```

```
-- =====
```

```
-- 0. user_credentials
```

```
CREATE TABLE user_credentials (
```

```

credential_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
password_hash VARCHAR(255),
authentication_type SMALLINT NOT NULL DEFAULT 1,
is_verified BOOLEAN NOT NULL DEFAULT FALSE,
two_factor_enabled BOOLEAN NOT NULL DEFAULT FALSE,
failed_login_count INT NOT NULL DEFAULT 0,
locked_until TIMESTAMPTZ,
last_password_change_at TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE INDEX idx_user_credentials_user ON user_credentials(user_id) WHERE deleted_at IS
NULL;

```

-- 7. user_profiles

```

CREATE TABLE user_profiles (
profile_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
national_id VARCHAR(255),
birth_date DATE,

```

```

avatar_url VARCHAR(۵۰۰),
gender SMALLINT,
address TEXT,
phone VARCHAR(۵۰),
description VARCHAR(۵۰۰),
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE INDEX idx_user_profiles_user ON user_profiles(user_id) WHERE deleted_at IS NULL;

```

-- V. tenant_users (پل ارتباطی مالیتی-تنت با لایه هویت)

ترکیب یکتا شدند tenant_id نکته حیاتی: فیلد ایمیل و نام کاربری به این لایه منتقل و با --

```

CREATE TABLE tenant_users (
tenant_user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
username VARCHAR(۱۰۰) NOT NULL,
tenant_user_email VARCHAR(۱۵۰) NOT NULL,
user_type SMALLINT NOT NULL DEFAULT ۱,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

```

```
created_by UUID,  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_tenant_users_username ON tenant_users(tenant_id, username)  
WHERE deleted_at IS NULL;
```

```
CREATE UNIQUE INDEX uq_tenant_users_email ON tenant_users(tenant_id, tenant_user_email)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_users_tenant ON tenant_users(tenant_id) WHERE deleted_at IS  
NULL;
```

```
CREATE INDEX idx_tenant_users_user ON tenant_users(user_id) WHERE deleted_at IS NULL;
```

-- 1. roles

```
CREATE TABLE roles (  
  role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  name VARCHAR(100) NOT NULL,  
  code VARCHAR(100) NOT NULL,  
  description VARCHAR(500),  
  role_type SMALLINT NOT NULL DEFAULT 1,  
  role_scope_type SMALLINT NOT NULL DEFAULT 1,  
  is_system_role BOOLEAN NOT NULL DEFAULT FALSE,  
  status SMALLINT NOT NULL DEFAULT 1,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```

created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE UNIQUE INDEX uq_roles_tenant_code ON roles(tenant_id, code) WHERE deleted_at IS
NULL;

CREATE INDEX idx_roles_tenant ON roles(tenant_id) WHERE deleted_at IS NULL;

-- =====
-- Phase 1 - Core ERP Foundation (REVISED) - PART 3 OF 4
-- Database Engine: PostgreSQL 14
-- =====

-- 9. permissions
CREATE TABLE permissions (
permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
code VARCHAR(100) NOT NULL,
name VARCHAR(200) NOT NULL,
module_name VARCHAR(100) NOT NULL,
group_name VARCHAR(100),
description VARCHAR(500),
permission_type SMALLINT NOT NULL DEFAULT 1,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

```



```
created_by UUID,  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_permissions_code ON permissions(code) WHERE deleted_at IS  
NULL;
```

```
CREATE INDEX idx_permissions_module ON permissions(module_name) WHERE deleted_at IS  
NULL;
```

```
-- 1. role_permissions
```

```
CREATE TABLE role_permissions (  
    role_permission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE RESTRICT,  
    permission_id UUID NOT NULL REFERENCES permissions(permission_id) ON DELETE RESTRICT,  
    access_level SMALLINT NOT NULL DEFAULT 1,  
    status SMALLINT NOT NULL DEFAULT 1,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    created_by UUID,  
    updated_at TIMESTAMPTZ,  
    updated_by UUID,  
    deleted_at TIMESTAMPTZ,  
    deleted_by UUID,  
    row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_role_permissions_composite ON role_permissions(role_id,
permission_id) WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_role_permissions_role ON role_permissions(role_id) WHERE deleted_at IS
NULL;
```

```
-- \\. user_roles
```

```
CREATE TABLE user_roles (
    user_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_user_id UUID NOT NULL REFERENCES tenant_users(tenant_user_id) ON DELETE
    RESTRICT,
    role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE RESTRICT,
    assigned_by UUID,
    start_date TIMESTAMPTZ,
    end_date TIMESTAMPTZ,
    status SMALLINT NOT NULL DEFAULT \,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT \
);
```

```
CREATE UNIQUE INDEX uq_user_roles_composite ON user_roles(tenant_user_id, role_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_user_roles_tenant_user ON user_roles(tenant_user_id) WHERE deleted_at IS
NULL;
```

```
CREATE INDEX idx_user_roles_role ON user_roles(role_id) WHERE deleted_at IS NULL;
```

```
-- ١٢. scopes
```

```
CREATE TABLE scopes (
```

```
  scope_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
  tenant_id UUID NOT NULL,
```

```
  scope_type SMALLINT NOT NULL,
```

```
  reference_type VARCHAR(١٠٠) NOT NULL,
```

```
  reference_id UUID NOT NULL,
```

```
  name VARCHAR(٢٠٠) NOT NULL,
```

```
  description VARCHAR(٥٠٠),
```

```
  status SMALLINT NOT NULL DEFAULT ١,
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
  created_by UUID,
```

```
  updated_at TIMESTAMPTZ,
```

```
  updated_by UUID,
```

```
  deleted_at TIMESTAMPTZ,
```

```
  deleted_by UUID,
```

```
  row_version BIGINT NOT NULL DEFAULT ١
```

```
);
```

```
CREATE INDEX idx_scopes_tenant ON scopes(tenant_id) WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_scopes_reference ON scopes(reference_type, reference_id) WHERE  
deleted_at IS NULL;
```

```
-- ١٣. user_scope_assignments
```

```
CREATE TABLE user_scope_assignments (
```

```

user_scope_assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_user_id UUID NOT NULL REFERENCES tenant_users(tenant_user_id) ON DELETE
RESTRICT,

scope_id UUID NOT NULL REFERENCES scopes(scope_id) ON DELETE RESTRICT,

is_primary BOOLEAN NOT NULL DEFAULT FALSE,

start_date TIMESTAMPTZ,

end_date TIMESTAMPTZ,

status SMALLINT NOT NULL DEFAULT \,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT \

);

CREATE UNIQUE INDEX uq_user_scope_assignment ON
user_scope_assignments(tenant_user_id, scope_id) WHERE deleted_at IS NULL;

CREATE INDEX idx_user_scope_assignment_user ON user_scope_assignments(tenant_user_id)
WHERE deleted_at IS NULL;

-- =====

-- Phase \ - Core ERP Foundation (REVISED) - PART 8 OF 8

-- Database Engine: PostgreSQL \A

-- =====

-- 18. organizational_unit_types

CREATE TABLE organizational_unit_types (

```

```

organizational_unit_type_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
code VARCHAR(50) NOT NULL,
name VARCHAR(100) NOT NULL,
description VARCHAR(500),
level_order SMALLINT NOT NULL DEFAULT 1,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE UNIQUE INDEX uq_organizational_unit_types_code ON organizational_unit_types(code)
WHERE deleted_at IS NULL;

```

```

CREATE INDEX idx_organizational_unit_types_status ON organizational_unit_types(status)
WHERE deleted_at IS NULL;

```

```
-- 10. organizational_units
```

```

CREATE TABLE organizational_units (
organizational_unit_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
company_id UUID NOT NULL REFERENCES erp_companies(company_id) ON DELETE RESTRICT,
parent_organizational_unit_id UUID REFERENCES organizational_units(organizational_unit_id)
ON DELETE RESTRICT,
organizational_unit_type_id UUID NOT NULL REFERENCES
organizational_unit_types(organizational_unit_type_id) ON DELETE RESTRICT,

```

```

code VARCHAR(۵۰) NOT NULL,
name VARCHAR(۲۰۰) NOT NULL,
manager_user_id UUID REFERENCES users(user_id) ON DELETE SET NULL, -- ارجاع منطقی مجاز بر
اساس اصول مرزبندی
description VARCHAR(۵۰۰),
level_order SMALLINT NOT NULL DEFAULT ۱,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE UNIQUE INDEX uq_organizational_units_company_code ON
organizational_units(company_id, code) WHERE deleted_at IS NULL;

CREATE INDEX idx_organizational_units_company ON organizational_units(company_id) WHERE
deleted_at IS NULL;

CREATE INDEX idx_organizational_units_parent ON
organizational_units(parent_organizational_unit_id) WHERE deleted_at IS NULL;

CREATE INDEX idx_organizational_units_type ON
organizational_units(organizational_unit_type_id) WHERE deleted_at IS NULL;

-- ۱۶. master_data_categories

CREATE TABLE master_data_categories (
master_data_category_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,

```

```

code VARCHAR(100) NOT NULL,
name VARCHAR(200) NOT NULL,
description VARCHAR(500),
is_system_category BOOLEAN NOT NULL DEFAULT FALSE,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE UNIQUE INDEX uq_master_data_categories_tenant_code ON
master_data_categories(tenant_id, code) WHERE deleted_at IS NULL;

```

```

CREATE INDEX idx_master_data_categories_tenant ON master_data_categories(tenant_id)
WHERE deleted_at IS NULL;

```

```
-- 1V. master_data_values
```

```

CREATE TABLE master_data_values (
    master_data_value_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL,
    master_data_category_id UUID NOT NULL REFERENCES
    master_data_categories(master_data_category_id) ON DELETE RESTRICT,
    parent_master_data_value_id UUID REFERENCES master_data_values(master_data_value_id)
    ON DELETE RESTRICT,
    code VARCHAR(100) NOT NULL,

```

```

name VARCHAR(۲۰۰) NOT NULL,
sort_order INT NOT NULL DEFAULT ۰,
extra_data JSONB, -- پشتیبانی از ویژگی‌های داینامیک انبار، کالا و شرکای تجاری
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE UNIQUE INDEX uq_master_data_values_category_code ON
master_data_values(master_data_category_id, code) WHERE deleted_at IS NULL;

CREATE INDEX idx_master_data_values_category ON
master_data_values(master_data_category_id) WHERE deleted_at IS NULL;

CREATE INDEX idx_master_data_values_extra_data_gin ON master_data_values USING GIN
(extra_data); -- JSONB برای کارایی فوق‌العاده بالا روی ویژگی‌های داینامیک GIN ایندکس

```

-- ۱۸. currencies

-- اصلاح استراتژیک بر اساس قوانین حاکمیت داده: کاتالوگ ارزها کاملاً عمومی و گلوبال (پلتفرمی) شد.

```

CREATE TABLE currencies (
currency_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
code VARCHAR(۱۰) NOT NULL,
name VARCHAR(۱۰۰) NOT NULL,
symbol VARCHAR(۲۰),
decimal_places SMALLINT NOT NULL DEFAULT ۲,

```



```

status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE UNIQUE INDEX uq_currencies_code ON currencies(code) WHERE deleted_at IS NULL;

```

```

-- ۱۹. tenant_currencies (جدول پیوند جدید مالیتی-تنتنت برای پایش ارز پایه هر شرکت بدون نشت داده)

CREATE TABLE tenant_currencies (
tenant_id UUID NOT NULL, -- ارجاع منطقی لایه‌ای به لایه ساس
currency_id UUID NOT NULL REFERENCES currencies(currency_id) ON DELETE RESTRICT,
is_base_currency BOOLEAN NOT NULL DEFAULT FALSE,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
PRIMARY KEY (tenant_id, currency_id)
);

CREATE INDEX idx_tenant_currencies_lookup ON tenant_currencies(tenant_id) WHERE
is_base_currency = TRUE;

```